PART II: KINETIC THEORY OF GASES

1 Oxygen gas having a volume of 1000 cm^3 at 40.0 ^oC and $1.01x10^{-5}$ Pa expands until its volume is 1500 cm^3 and its pressure is $1.06x10^{-5}$ Pa.

Find (a) the number of moles of oxygen present and (b) the final temperature of the sample.

2 A quantity of ideal gas at 10.0^oC and 100 kPa occupies a volume of 2.50 m^3.

(a) How many moles of the gas are present?

(b) If the pressure is now raised to 300 kPa and the temperature is raised to 30.0^oC, how much volume does the gas occupy? Assume no leaks.

3 The best laboratory vacuum has a pressure of about $1.00x10^{-18}$ atm, or $1.01x10^{-13}$ Pa. How many gas molecules are there per cubic centimeter in such a vacuum at 293 K?

4 Suppose 1.80 mol of an ideal gas is taken from a volume of 3.00 m^3 to a volume of 1.50 m^3 via an isothermal compression at 30^oC.

(a) How much energy is transferred as heat during the compression?

(b) Is the transfer to or from the gas?

5 Submarine rescue. When the U.S. submarine Squalus became disabled at a depth of 80 m, a cylindrical chamber was lowered from a ship to rescue the crew. The chamber had a radius of 1.00 m and a height of 4.00 m, was open at the bottom, and held two rescuers. It slid along a guide cable that a diver had attached to a hatch on the submarine. Once the chamber reached the hatch and clamped to the hull, the crew could escape into the chamber. During the descent, air was released from tanks to prevent water from flooding the chamber. Assume that the interior air pressure matched the water pressure at depth h as given by $p_0 + pgh$, where p_0 = 1.000 atm is the surface pressure and r = 1024 kg/m^3 is the density of seawater. Assume a surface temperature of 20.0^0C and a submerged water temperature of -30.0^0C. (a) What is the air volume in the chamber at the surface? (b) If air had not been released from the tanks, what would have been the air volume in the chamber at depth h = 80.0 m? (c) How many moles of air were needed to be released to maintain the original air volume in the chamber?

6 A sample of an ideal gas is taken through the cyclic process *abca* shown in **Fig. below**. The scale of the vertical axis is set by p_b = 7.5 kPa and p_{ac} = 2.5 kPa. At point *a*, T = 200 K.

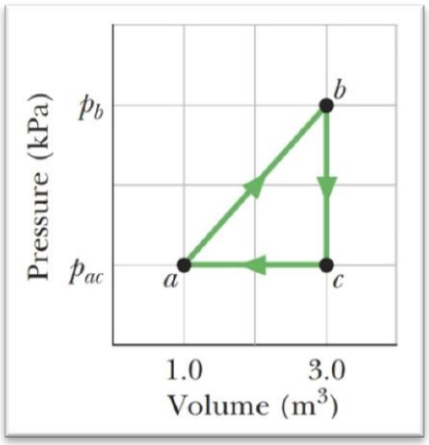
(a) How many moles of gas are in the sample?

What are:

(b) the temperature of the gas at point *b*?

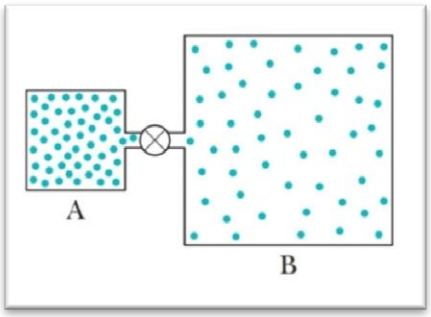
(c) the temperature of the gas at point *c*?

(d) the net energy added to the gas as heat during the cycle?



7 An air bubble of volume 20 cm^3 is at the bottom of a lake 40 m deep, where the temperature is 4.0^0C. The bubble rises to the surface, which is at a temperature of 20^0C. Take the temperature of the bubble's air to be the same as that of the surrounding water. Just as the bubble reaches the surface, what is its volume?

8 Container A in **Fig. below** holds an ideal gas at a pressure of 5.0 x 10^5 P_a and a temperature of 300 K. It is connected by a thin tube (and a closed valve) to container B, with four times the volume of A. Container B holds the same ideal gas at a pressure of 1.0 x 10^5 P_a and a temperature of 400 K. The valve is opened to allow the pressures to equalize, but the temperature of each container is maintained. What then is the pressure?



9 (a) Compute the rms speed of a nitrogen molecule at 20.0^0C. The molar mass of nitrogen molecules (N_2) is given in **Table below**. At what temperatures will the rms speed be (b) half that value and (c) twice that value?

Table 19-1 Some RMS Speeds at R Temperature ($T = 300$ K)[a]

Gas	Molar Mass (10^{-3} kg/mol)	v_{rms} (m/s)
Hydrogen (H_2)	2.02	1920
Helium (He)	4.0	1370
Water vapor (H_2O)	18.0	645
Nitrogen (N_2)	28.0	517
Oxygen (O_2)	32.0	483
Carbon dioxide (CO_2)	44.0	412
Sulfur dioxide (SO_2)	64.1	342

[a]For convenience, we often set room temperature equal to 300 K even though (at 27°C or 81°F) that represents a fairly warm room.

10 At 273 K and 1.00 x 10^{-2} atm, the density of a gas is 1.24 x 10^{-5} g/cm^3.

(a) Find v_{rms} for the gas molecules.

(b) Find the molar mass of the gas and

(c) identify the gas. See Table above.

11 Determine the average value of the translational kinetic energy of the molecules of an ideal gas at temperatures (a) 0.00°C and (b) 100°C. What is the translational kinetic energy per mole of an ideal gas at (c) 0.00^{0}C and (d) 100^{0}C?

12 Water standing in the open at 32.0°C evaporates because of the escape of some of the surface molecules. The heat of vaporization (539 cal/g) is approximately equal to εn, where ε is the average energy of the escaping molecules and n is the number of molecules per gram. (a) Find ε. (b) What is the ratio of ε to the average kinetic energy of H_2O molecules, assuming the latter is related to temperature in the same way as it is for gases?

13 At what frequency would the wavelength of sound in air be equal to the mean free path of oxygen molecules at 1.0 atm pressure and 0.00^{0}C? The molecular diameter is 3.0 x 10^{-8} cm.

14 In a certain particle accelerator, protons travel around a circular path of diameter 23.0 m in an evacuated chamber, whose residual gas is at 295 K and 1.00 x 10^{-6} torr pressure. (a) Calculate the number of gas molecules per cubic centimeter at this pressure. (b) What is the mean free path of the gas molecules if the molecular diameter is 2.0 x 10^{-8} cm?

15 At 20°C and 750 torr pressure, the mean free paths for argon gas (Ar) and nitrogen gas (N_2) are λ_{Ar} = 9.9 x 10^6 cm and λ_{N2} = 27.5 x 10^6 cm. (a) Find the ratio of the diameter of an Ar atom to that of an N_2 molecule. What is the mean free path of argon at (b) 20^0C and 150 torr, and (c) -40°C and 750 torr?

16 The speeds of 22 particles are as follows (N_i represents the number of particles that have speed v_i):

N_i	2	4	6	8	2
V_i(cm/s)	1.0	2.0	3.0	4.0	5.0

What are (a) v_{avg}, (b) v_{rms}, and (c) v_P?

17 The most probable speed of the molecules in a gas at temperature T_2 is equal to the rms speed of the molecules at temperature T_1. Find T_2 / T_1.

MCQs

1. The quantity PV/kT (k = Boltzmann's constant) represents:

a. Number of moles of the gas
b. Total mass of the gas
c. Number of molecules in the gas
d. Density of the gas

2. In the process PV = constant, the pressure P versus density graph ρ of an ideal gas is:

a. A parabola
b. A straight line passing through origin
c. A straight line parallel to ρ-axis
d. A straight line parallel to P-axis

3. The V_{rms} of gas molecules is 300 m/sec. If its absolute temperature is reduced to half and molecular weight is doubled, the V_{rms} will become:

a. 75 m/s
b. 150 m/s
c. 300 m/s
d. 600 m/s

4. Average kinetic energy of molecules is:

a. Directly proportional to square root of temperature
b. Directly proportional to absolute temperature
c. Independent of absolute temperature

d. Inversely proportional to absolute temperature

5. The specific heat of a gas in an isothermal process is

a. Zero

b. Negative

c. Remains constant

d. Infinite

6. The specific heat of a gas in an adiabatic process is

a. Zero

b. Negative

c. Remains constant

d. Infinite

7. An ideal gas is taken in a sealed container at 300 K. it is heated at constant volume to a temperature 600 K. the mean kinetic energy of its molecules is

a. Halved

b. Doubled

c. Tripled

d. Quadrupled

8. If 70 calories heat is required to raise the temperature of 2 moles of an ideal diatomic gas at constant pressure from 30°C to 35°C. the amount of heat required to raise the temperature of the same gas through the same range at constant volume is

a. 30 cal

b. 50 cal

c. 70 cal

d. 90 cal

9. Which of the following is the unit of specific heat?

a. J kg/°C

b. J/kg°C

c. kg °C/J

d. J kg/°C^2

10. The mean free path of a gas molecule is λ_1 at temperature T. The pressure and temperature of the gas are made two times their original value. The new men free path λ_2 is:

a. Two times of λ_1

b. Equal to λ_1

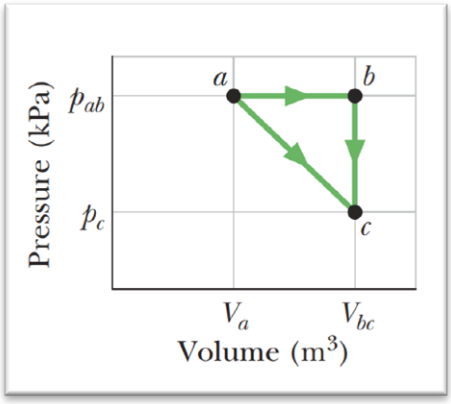
c. Half of λ_1

d. Four times of λ_1

18 A hydrogen molecule (diameter 1.0 x 10^{8} cm), traveling at the rms speed, escapes from a 4000K furnace into a chamber containing cold argon atoms (diameter 3.0 x 10^{8} cm) at a density of 4.0 x 10^{19} atoms/cm^{3}. (a) What is the speed of the hydrogen molecule? (b) If it collides with an argon atom, what is the closest their centers can be, considering each as spherical? (c) What is the initial number of collisions per second experienced by the hydrogen molecule?

19 What is the internal energy of 1.0 mol of an ideal monatomic gas at 273 K?

20 One mole of an ideal diatomic gas goes from a to c along the diagonal path **in Fig. below**. The scale of the vertical axis is set by p_{ab} = 5.0 kP$_a$ and p_c = 2.0 kP$_a$, and the scale of the horizontal axis is set by V_{bc} = 4.0 m^3 and V_a = 2.0 m^3. During the transition, (a) what is the change in internal energy of the gas, and (b) how much energy is added to the gas as heat? (c) How much heat is required if the gas goes from a to c along the indirect path *abc*?



21 A container holds a mixture of three nonreacting gases: 2.40 mol of gas 1 with C_{V1} 12.0 J/mol·K 1.50 mol of gas 2 with C_{V2} =12.8 J/mol·K, and 3.20 mol of gas 3 with C_{V3} = 20.0 J/mol·K. What is C_V of the mixture?

22 We give 70 J as heat to a diatomic gas, which then expands at constant pressure. The gas molecules rotate but do not oscillate. By how much does the internal energy of the gas increase?

23 Suppose 12.0 g of oxygen (O_2) gas is heated at constant atmospheric pressure from 25.0^{0}C to 125^{0}C. (a) How many moles of oxygen are present? (See Table below for the molar mass.) (b) How much energy is transferred to the oxygen as heat? (The molecules rotate but do not oscillate.) (c) What fraction of the heat is used to raise the internal energy of the oxygen?

Table 19-1 Some RMS Speeds at Rc Temperature ($T = 300$ K)[a]

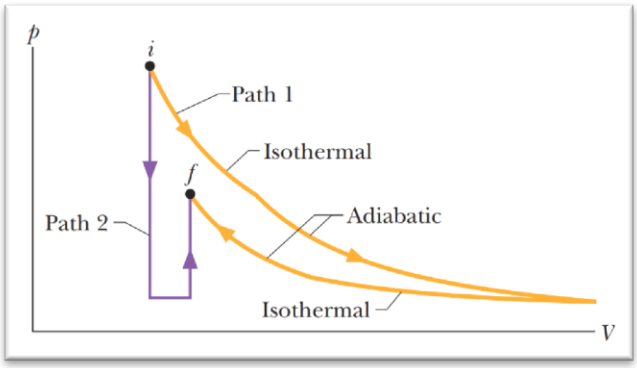
Gas	Molar Mass (10^{-3} kg/mol)	v_{rms} (m/s)
Hydrogen (H_2)	2.02	1920
Helium (He)	4.0	1370
Water vapor (H_2O)	18.0	645
Nitrogen (N_2)	28.0	517
Oxygen (O_2)	32.0	483
Carbon dioxide (CO_2)	44.0	412
Sulfur dioxide (SO_2)	64.1	342

[a]For convenience, we often set room temperature equal to 300 K even though (at 27°C or 81°F) that represents a fairly warm room.

24 Suppose 1.00 L of a gas with γ = 1.30, initially at 273 K and 1.00 atm, is suddenly compressed adiabatically to half its initial volume. Find its final (a) pressure and (b) temperature. (c) If the gas is then cooled to 273 K at constant pressure, what is its final volume?

25 *Opening champagne*. In a bottle of champagne, the pocket of gas (primarily carbon dioxide) between the liquid and the cork is at pressure of *pi* = 5.00 atm. When the cork is pulled from the bottle, the gas undergoes an adiabatic expansion until its pressure matches the ambient air pressure of 1.00 atm. Assume that the ratio of the molar specific heats is $\gamma = \frac{4}{3}$. If the gas has initial temperature T_i = 5.00^{0}C, what is its temperature at the end of the adiabatic expansion?

26 Figure below shows two paths that may be taken by a gas from an initial point *i* to a final point *f*. Path 1 consists of an isothermal expansion (work is 50 J in magnitude), an adiabatic expansion (work is 40 J in magnitude), an isothermal compression (work is 30 J in magnitude), and then an adiabatic compression (work is 25 J in magnitude). What is the change in the internal energy of the gas if the gas goes from point *i* to point *f* along path 2?



27 An ideal diatomic gas, with rotation but no oscillation, undergoes an adiabatic compression. Its initial pressure and volume are 1.20 atm and 0.200 m^3. Its final pressure is 2.40 atm. How much work is done by the gas?

28 Figure below shows a cycle undergone by 1.00 mol of an ideal monatomic gas. The temperatures are T_1 = 300 K, T_2 = 600 K, and T_3 = 455 K. For 1 : 2, what are (a) heat Q, (b) the change in internal energy ΔE_{int}, and (c) the work done W? For 2 : 3, what are (d) Q, (e) ΔE_{int}, and (f) W? For 3 : 1, what are (g) Q, (h) ΔE_{int}, and (i) W? For the full cycle, what are (j) Q, (k) ΔE_{int}, and (l) W? The initial pressure at point 1 is 1.00 atm (= 1.013 x 10^5 Pa). What are the (m) volume and (n) pressure at point 2 and the (o) volume and (p) pressure at point 3?

